

codes.R

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```
# Tue May 14 13:58:44 2024 -----
#####
###           Installing packages from the r cran archive           #####
###                                           #####
#####
### dependencies = true implies you're interested in the either dependency else you're not

### Installing packages no more on r cran (zelifg)
### now there some dependencies so you need to install them first

### Obtaining data or functions from a package
### 1. load the package using library function
### 2.

library(AER)

## Loading required package: car

## Loading required package: carData

## Loading required package: lmtest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric

## Loading required package: sandwich

## Loading required package: survival
```

```
?BankWages
```

```
## starting httpd help server ...
```

```
## done
```

```
data("BankWages")
```

```
# Bankwages cannot just be loaded without the data function
```

```
?anorexia # Error because the required package is not loaded
```

```
## No documentation for 'anorexia' in specified packages and libraries:
```

```
## you could try '??anorexia'
```

```
??anorexia # Search through all the installed packages
```

```
### Attaching a data from a package
```

```
data("BankWages")
```

```
#####  
### DATA STRUCTURES #####  
#####
```

```
### 1. VECTOR
```

```
### 2. MATRIX : contains only one data types, two dimensional
```

```
### 3. DATAFRAME : can contain different data tptes, two dimensional
```

```
### 2 and 3 are the same in terms of structure
```

```
### 4. LIST: can contain other data structures and one dimensional
```

```
#####  
### DATA TYPES #####  
#####
```

```
### 1. Numeric
```

```
### 2. Character
```

```
### 3. Missing data (NA)
```

```
### 4. Logical (TRUE/FALSE)
```

```
### 5. Factor: Norminal and ordinal
```

```
### 6. Date and Time
```

```
### 1. VECTOR
```

```
# we use the concatenation function, c to create a vector in r
```

```
Age <- c(28,18,23,20,21)
```

```
Religion <- c("christian","muslim","traditionalist","other","christian")
```

```
Weight <- c(70,150,65,85,92)
```

```
Height <- c(1.65,1.71,1.82,1.79,1.54)
```

```
None <- c(NA,NA)
```

```
T_F_Data <- c(FALSE,TRUE,T,F,T)
```

```
10:1 # Numbers from 10 to 1 in steps of 1
```

```
## [1] 10 9 8 7 6 5 4 3 2 1
```

```
rep(3,10) # Repete 3 10 times
```

```
## [1] 3 3 3 3 3 3 3 3 3 3
```

```
seq(from=1,to=100,by=10)
```

```
## [1] 1 11 21 31 41 51 61 71 81 91
```

```
### 2. MATRIX
```

```
Data <- 1:18
```

```
?matrix
```

```
Mat.Data <- matrix(data = Data, nrow = 3, ncol = 6, byrow = FALSE)
```

```
Mat.Data <- matrix(data = Data, nrow = 3, ncol = 6, byrow = TRUE)
```

```
cbind(Age,Weight,Height)
```

```
##      Age Weight Height
## [1,]  28     70   1.65
## [2,]  18    150   1.71
## [3,]  23     65   1.82
## [4,]  20     85   1.79
## [5,]  21     92   1.54
```

```
rbind(Age,Weight,Height)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## Age    28.00 18.00 23.00 20.00 21.00
## Weight 70.00 150.00 65.00 85.00 92.00
## Height 1.65  1.71 1.82 1.79 1.54
```

```
### 3. DATAFRAMES
```

```
rbind(Age,Weight,Height,Religion)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## Age      "28"      "18"      "23"      "20"      "21"
## Weight    "70"      "150"     "65"      "85"      "92"
## Height    "1.65"     "1.71"     "1.82"     "1.79"     "1.54"
## Religion  "christian" "muslim" "traditionalist" "other" "christian"
```

```
# Now everything has been changed to character.... that's why we use data/frame function
```

```
Df.Data <- data.frame(Age,Weight,Height,Religion)
```

```
### 4. LIST
```

```
mylist <- list(Numeric = Age,Character = Religion,Matrix = Mat.Data, Dataframe = Df.Data)
```

```
### Retrieving or subscri[ption observations from data structures
```

```
### 1. vectors
```

```
Age[3]
```

```
## [1] 23
```

```
Religion[3]
```

```
## [1] "traditionalist"
```

```
### Using the object names  
names(Age) <- LETTERS[1:5]  
Age["C"]
```

```
## C
```

```
## 23
```

```
### 2. Matrix and data frames  
### You have rows and columns  
MAt.Data[2,3]
```

```
## [1] 9
```

```
MAt.Data[2,];MAt.Data[2,1:6]
```

```
## [1] 7 8 9 10 11 12
```

```
## [1] 7 8 9 10 11 12
```

```
### For taking observations for a variable  
Df.Data$Age
```

```
## [1] 28 18 23 20 21
```

```
### 4. List  
# just as vectors  
mylist[1] # comes as a list
```

```
## $Numeric
```

```
## [1] 28 18 23 20 21
```

```
mylist$Numeric #comes
```

```
## [1] 28 18 23 20 21
```